DHARSAN R

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SUMMARY

Self-taught full-stack developer, pivoted from a Mechanical Engineering background into backend and full-stack software development. Built a full-stack group-expense ledger application with end-to-end ownership, from database schema design to deployed UI. The project features OAuth authentication, a custom debt-simplification algorithm, and optimized multi-table PostgreSQL queries. Hands-on experience with TypeScript, React, PostgreSQL, and Express.

EDUCATION

Sona college of Technology Tamil Nadu, Salem | 2022 - 2026

- Bachelor of Engineering in Mechanical - 86%

PSG Matriculation Higher Secondary School Tamil Nadu, Sankari | 2021 - 2022

- Higher Secondary Education -78.6%

SKILLS

Languages & Frameworks	React 19, Express.js, SQL, TypeScript, JavaScript, Node.js,
Core Concepts	Object-Oriented Programming (OOP), Data Structures & Algorithms
Backend & Infrastructure	REST API, OAuth 2.0(Supabase Auth), PostgreSQL, Supabase.
Tooling & UI	Git, Shadcn UI, Tailwind CSS.

PROJECTS

Expense Splitter Web App | React (Vite), Node.js, Express, TypeScript, PostgreSQL, Drizzle ORM, Supabase, Tailwind CSS July 2026 - August

- Architected full-stack group ledger with modular Express routers and TypeScript strict mode, secured with Supabase OAuth 2.0 (Google login) and JWT tokens.
- Implemented a greedy transaction-simplification algorithm that resolved complex group debts into the minimum number of transfers (e.g., 12-person group: 15 transfers → 5 transfers).
- Optimized PostgreSQL with window functions and CTEs to compute net balances per person in a single query, eliminating N+1 issues and redundant database round-trips.
- Enforced atomic transactions across multi-table expense and split operations to guard against partial writes on runtime faults or concurrent updates.
- Migrated from raw pg queries to Drizzle ORM for type-safe database operations,eliminates runtime type errors.
- Added input validation with Zod schemas and custom middleware to guard against injection attacks and malformed requests at the server boundary.
- Built responsive UI using Shadcn UI and Tailwind CSS with debounced autocomplete search to minimize redundant API calls.